

⇒ 8 puzzle problem.

DFS (start, goal)

create an empty stack

create an empty visited set

Push start into stack

while stack is not empty:

pop the top state

if state == goal:

return solution

if state is not visited:

mark state as visited

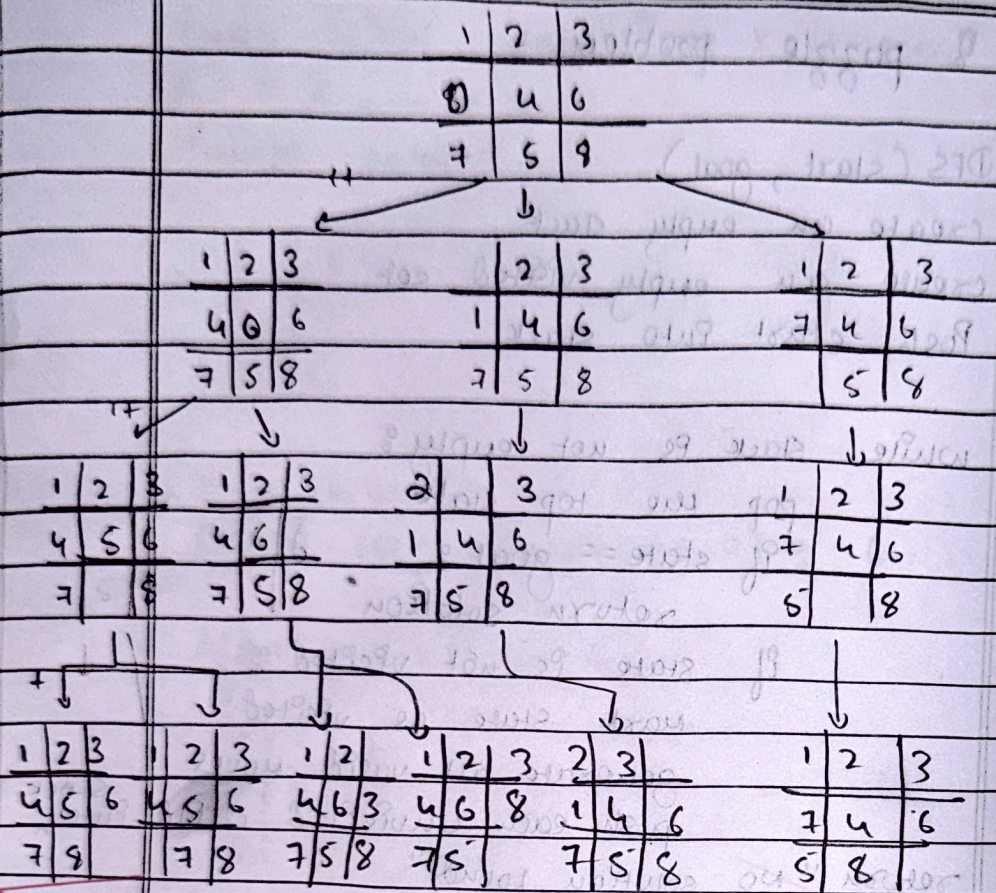
generate all valid moves

push each unvisited child into ^{stack}

return "no solution found"

1	2	9
4	5	6
7	8	0

Page No.			
Date.			



Good

path cost = 3

Good

--- Define Initial state ---

Enter 9 numbers (0-8) separated by spaces (0 represents the blank space)
 >> 1 2 3 0 4 6 7 5 8

--- Define Goal state ---

Enter 9 numbers (0-8) separated by spaces (0 represents the blank space)
 >> 1 2 3 4 5 6 7 8 0

DFS Pathway #1 (Path cost: 3)
 Initial setup

1 2 3

4 6

7 5 8

Step 1: move empty spot 'right'

1 2 3

4 - 6

7 5 8

Step 2: move empty spot 'down'

1 2 3

4 5 6

7 - 8

~~Step 3: move empty spot 'right'~~

~~1 2 3~~

~~4 5 6~~

~~7 8 -~~

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